

		$= 3.0 \text{ J}$
11	D	Speed of the satellite $v = r\omega$ $= r \left(\frac{2\pi}{T} \right)$ $= \left[(6400 + 36000) \times 10^3 \right] \left(\frac{2\pi}{24 \times 60 \times 60} \right)$ $= 3083$ $\approx 3100 \text{ m s}^{-1}$ <i>Note: A common mistake is to forget to add on the value of the Earth to the height of the satellite above the surface of the Earth.</i>

No.	Answer	Solution
12	C	Use $nV = nRT$